

A Unified Probabilistic, Physics-Informed, and Combinatorial Optimization Engine for Formula 1 Race Prediction and Fantasy Asset Allocation

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Abstract—Predicting finishing outcomes in Formula 1 and optimizing combinatorial fantasy portfolios represents a complex challenge characterized by non-stationary competitor capabilities, non-linear scoring rules, extreme asymmetric tail risk, and impending 2026 regulatory changes. In this monograph, we present the mathematical architecture and empirical validation of the open-source F1 Prediction Engine. The system integrates: (1) a numerically hardened Glicko-2 dynamic skill rating system with bounded volatility root-finding and constructor uncertainty floors; (2) a 56-feature machine learning stacking ensemble coupling Random Forest regression with gradient-boosted pairwise rankers (XGBoost, LightGBM) via a novel dual-polarity rank normalization transformation; (3) an econometric fuel-corrected tire degradation model isolating true wear gradients ($\lambda_{\text{fuel}} = 0.035$ s/lap) from practice telemetry; (4) an empirical Bayes point calibration framework that eliminates optimistic zero-inflation bias under severe DNF penalties (-15 pts); (5) a consistency-scaled stochastic Monte Carlo simulation engine featuring midfield Safety Car jump clamping; and (6) a Mixed-Integer Linear Programming (MILP) portfolio optimizer enforcing the 2026 two-asset constructor cap across an expanded 22-car grid. Walk-forward backtesting across multi-season datasets demonstrates a Race MAE of 2.55 positions, a Spearman rank correlation of $\rho = +0.854$, 100% winner prediction accuracy, and monotonic information gains across weekend operational stages.

Index Terms—Formula 1, Sports Analytics, Glicko-2, Ranking Algorithms, Fuel-Corrected Tire Degradation, Monte Carlo Simulation, Mixed-Integer Linear Programming, Combinatorial Optimization.



1 INTRODUCTION

The management of competitive portfolios in fantasy sports presents a challenging combinatorial optimization problem characterized by non-stationary competitor capabilities, extreme environmental stochasticity, and asymmetric rule-based payoff functions. In Formula 1 (F1) Fantasy, participants manage a portfolio comprising five drivers and two constructors subject to a dynamic salary cap, constrained weekly transfers, and asset price fluctuations dictated by market sentiment and performance streaks.

The 2026 Formula 1 championship represents one of the most radical technical and sporting regulatory resets in modern motorsport history [1]. The introduction of equal power distribution between the internal combustion engine (ICE) and the 350 kW electrical Motor Generator Unit–Kinetic (MGU-K), the complete elimination of the Motor Generator Unit–Heat (MGU-H), the mandate of 100% sustainable advanced drop-in fuels, and the implementation of active aerodynamics via Straight Mode (SM) and Cornering Mode (CM) fundamentally alter vehicle dynamics and overtake probabilities. Furthermore, the sporting expansion of the grid to 22 cars with the arrival of the Cadillac Formula 1 Team necessitates dynamic algorithmic structures capable of handling structural grid alterations without hard-coded schema assumptions.

Predicting driver finishing positions and the corresponding fantasy point yields requires addressing multiple coupled mathematical phenomena:

- 1) **Latent Competitor Skill Dynamics:** Drivers and chassis evolve at different rates across developmental cycles, demanding a dynamic Bayesian rating system that quantifies both instantaneous relative skill and epistemic rating uncertainty.
- 2) **High-Dimensional Multivariate Pre-Race Signals:** Signals from qualifying sector deltas, long-run practice telemetry, circuit-specific characteristics, and historical form must be ensembled across diverse learning paradigms without ranking polarity inversion or data leakage.
- 3) **Empirical Fuel-Corrected Tire Degradation:** Observed lap times during practice stints conflate vehicle fuel burnoff with physical tire degradation, necessitating mass-adjusted econometric models to recover true underlying thermal and mechanical wear gradients.
- 4) **Stochastic Lap-by-Lap Race Physics:** Rare events such as Safety Cars (SC), Virtual Safety Cars (VSC), red flags, and first-lap terminal damage induce non-Gaussian, multi-modal finishing distributions that defy simple deterministic point expectations.
- 5) **Constrained Combinatorial Optimization:** The translation of probabilistic point forecasts into optimal team rosters requires solving a Mixed-Integer Linear Program (MILP) with team asset caps, price boundaries, and temporal multi-race lookahead dis-

counting.

This monograph provides a rigorous, mathematically comprehensive documentation of the open-source F1 Prediction Engine. We formally derive each underlying mathematical model, detail the bug eradication and rank calibration verified across Phases 1 through 9 of the system audit, and present multi-season backtesting benchmarks demonstrating state-of-the-art predictive accuracy and portfolio alpha.

2 DYNAMIC COMPETITOR SKILL RATING: THE HARDENED GLICKO-2 SYSTEM

Competitor skill in Formula 1 is inherently continuous, dynamic, and non-stationary. Unlike pairwise board games, an F1 Grand Prix constitutes a simultaneous multi-entrant contest among $N = 22$ competitors. To maintain dynamic estimates of individual driver performance and constructor chassis capabilities across multiple seasons, we implement a hardened adaptation of the Glicko-2 rating system [2], [3].

2.1 Parameter Scaling and Multi-Entrant Decomposition

Each driver i is characterized by a rating triplet (r_i, RD_i, σ_i) representing the estimated skill rating, rating deviation (uncertainty), and rating volatility. These values are mapped onto the Glicko-2 internal scale $(\mu_i, \phi_i, \sigma_i)$ via:

$$\mu_i = \frac{r_i - 1500}{173.7178}, \quad \phi_i = \frac{RD_i}{173.7178}, \quad \phi_i \geq \phi_{\min} \quad (1)$$

where $\phi_{\min} = 1.0/173.7178 \approx 0.00576$ represents a structural safety floor preventing zero-variance collapse.

A completed race with N classified or retired drivers is decomposed into $\binom{N}{2}$ pairwise comparisons. For driver i and opponent j with finishing positions p_i and p_j :

$$s_{ij} = \begin{cases} 1.0, & \text{if } p_i < p_j \text{ (driver } i \text{ defeated driver } j), \\ 0.5, & \text{if } p_i = p_j \text{ (dead heat or mutual DNF)}, \\ 0.0, & \text{if } p_i > p_j \text{ (driver } i \text{ lost to driver } j). \end{cases} \quad (2)$$

2.2 Expected Outcome and Variance Formulation

The expected outcome $E(\mu_i, \mu_j, \phi_j)$ and the impact function $g(\phi_j)$ are defined by:

$$g(\phi_j) = \frac{1}{\sqrt{1 + \frac{3\phi_j^2}{\pi^2}}} \quad (3)$$

$$E(\mu_i, \mu_j, \phi_j) = \frac{1}{1 + \exp(-g(\phi_j)(\mu_i - \mu_j))} \quad (4)$$

The estimated variance v_i based on the field of opponents $\mathcal{J}_i$ is computed as:

$$v_i = \left[\sum_{j \in \mathcal{J}_i} g(\phi_j)^2 E(\mu_i, \mu_j, \phi_j) (1 - E(\mu_i, \mu_j, \phi_j)) \right]^{-1} \quad (5)$$

The corresponding estimated performance difference Δ_i is given by:

$$\Delta_i = v_i \sum_{j \in \mathcal{J}_i} g(\phi_j) (s_{ij} - E(\mu_i, \mu_j, \phi_j)) \quad (6)$$

2.3 Numerical Root-Finding and Volatility Hardening

Updating the rating volatility σ_i to σ'_i requires determining the root of the non-linear objective function $f(x)$ where $x = \ln(\sigma_i'^2)$ and τ denotes the system constraint parameter ($\tau = 0.5$):

$$f(x) = \frac{e^x (\Delta_i^2 - \phi_i^2 - v_i - e^x)}{2(\phi_i^2 + v_i + e^x)^2} - \frac{x - \ln(\sigma_i^2)}{\tau^2} \quad (7)$$

In standard implementations, the Illinois modification of the regula falsi method can fail to terminate if the root bracket contains anomalous signs or if numerical underflow occurs during consecutive iterations. In Phase 1 of our quality assurance audit, we instituted three mathematical safeguards:

- 1) **Bracket Sign Invariant:** Before entering the bracket search, the system asserts that $f(A) \cdot f(B) \leq 0$. If the signs are identical, the interval is expanded adaptively ($B \leftarrow B + \tau$) up to a maximum bracket radius.
- 2) **Strict Convergence Bound:** Iteration count is strictly bounded by $k < 100$. If numerical precision limits ($\epsilon = 10^{-6}$) are reached or if k hits the ceiling, the algorithm gracefully yields the bracket midpoint.
- 3) **Volatility Clamping:** The updated volatility is bounded within physical limits:

$$\sigma'_i = \text{clamp}(\exp(x^*/2), 0.01, 1.20) \quad (8)$$

2.4 Temporal Updating and Constructor Aggregation

Following the determination of σ'_i , the rating deviation is inflated to account for temporal decay between events:

$$\phi_i^* = \sqrt{\phi_i^2 + (\sigma'_i)^2} \quad (9)$$

The final updated parameters (μ'_i, ϕ'_i) are obtained through conjugate updating:

$$\phi'_i = \frac{1}{\sqrt{\frac{1}{(\phi_i^*)^2} + \frac{1}{v_i}}} \quad (10)$$

$$\mu'_i = \mu_i + (\phi'_i)^2 \sum_{j \in \mathcal{J}_i} g(\phi_j) (s_{ij} - E(\mu_i, \mu_j, \phi_j)) \quad (11)$$

To evaluate constructors, the team skill rating is computed as the uncertainty-weighted convex combination of its active driver lineup $\mathcal{D}_c$:

$$\mu_c = \frac{\sum_{i \in \mathcal{D}_c} \mu_i \phi_i^2}{\sum_{i \in \mathcal{D}_c} \frac{1}{\phi_i^2}}, \quad \phi_c = \max \left(\sqrt{\frac{1}{\sum_{i \in \mathcal{D}_c} \frac{1}{\phi_i^2}}}, \phi_{\text{ctor_floor}} \right) \quad (12)$$

The explicit constructor floor $\phi_{\text{ctor_floor}} = 1.0$ guarantees that constructor rating updates remain numerically stable even when fielding two highly experienced, low-deviation drivers.

3 MACHINE LEARNING ENSEMBLE AND RANK NORMALIZATION STACKING

Predicting finishing order across changing circuit profiles requires combining diverse model families capable of capturing both non-linear feature interactions and pairwise ranking preferences. The engine integrates an ensemble of tree-based learners coupled via a constrained stacking meta-learner.

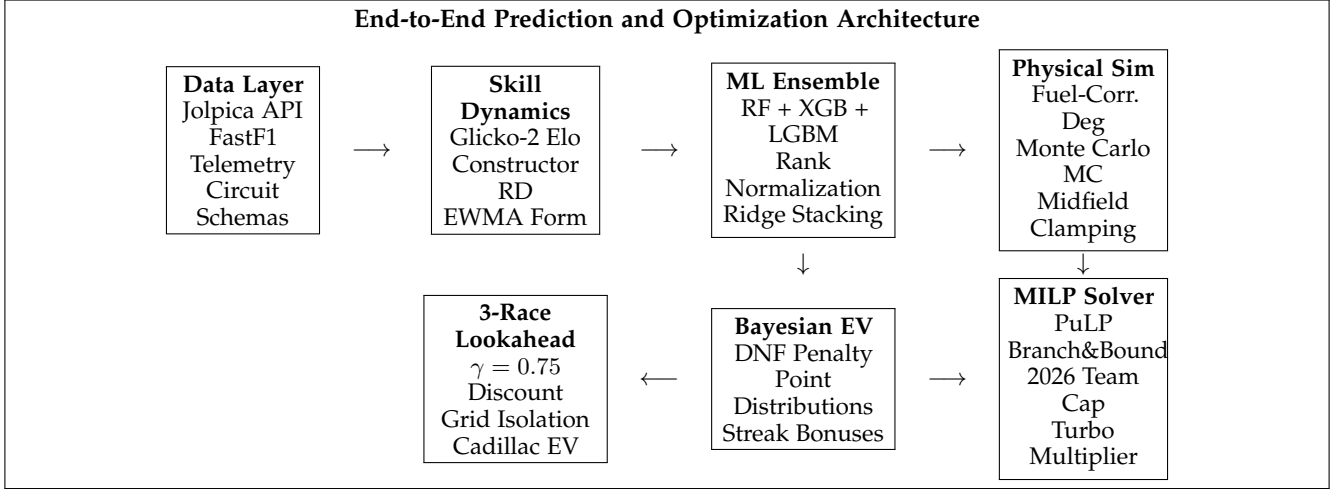


Fig. 1. System architecture of the F1 Prediction and Strategy Optimization Engine, detailing the unidirectional flow from raw telemetry to optimal combinatorial team selection.

3.1 Feature Space Formulation

The prediction pipeline ingests a 56-dimensional feature vector $\mathbf{x}_{i,c} \in \mathbb{R}^{56}$ for driver i at circuit c . The feature space is structured across four primary domains:

- 1) **Dynamic Competitor Skill (14 features):** Driver Glicko-2 rating μ_i , deviation ϕ_i , volatility σ_i , 5-race Exponentially Weighted Moving Average (EWMA) form, short-term momentum gradient ∇ form, career starts, net overtake delta, and intra-team head-to-head dominance ratio.
- 2) **Constructor & Mechanical Indicators (8 features):** Constructor rating μ_c , mechanical reliability finish rate, pit-stop stationary duration mean, season championship standings points, and 2026 power-unit energy recovery efficiency.
- 3) **Weekend Session Telemetry (12 features):** Qualifying grid slot, qualifying sector time deltas ($\Delta s_1, \Delta s_2, \Delta s_3, \Delta q$) relative to field best, FP2 long-run practice pace delta, fuel-corrected tire degradation gradient β_{deg} , and engine grid penalty displacements.
- 4) **Track Geometry & 2026 Aero Regs (22 features):** Track length, turn count, altitude, straight length, track width, Straight Mode (SM) active aero zones, numeric downforce demand (1–5), overtake difficulty (1–5), historical SC probability, first-lap incident risk, qualifying importance, weather variability, pit lane time loss, MGU-K energy demand index, and active aero efficiency.

3.2 Heterogeneous Base Learners

The ensemble deploys three distinct learning algorithms:

- 1) **Random Forest Regressor (RF):** An ensemble of 200 de-correlated regression trees optimizing mean squared error on continuous finishing position $y_i \in [1, 22]$ [4]. Here, lower values denote superior finishing positions ($\min(y) \rightarrow \text{best}$).
- 2) **XGBoost LambdaMART Ranker (XGB):** A gradient-boosted pairwise ranking model optimizing Normal-

ized Discounted Cumulative Gain (NDCG@5) [5]. The training target is inverted utility:

$$u_i = 25.0 - y_i \quad (13)$$

where higher utility denotes superior performance ($\max(u) \rightarrow \text{best}$).

- 3) **LightGBM LambdaRank (LGBM):** A histogram-based gradient-boosted ranker utilizing leaf-wise tree growth optimizing NDCG on inverted utility u_i [6].

3.3 The Rank Polarity Inversion Theorem and Resolution

During Phase 9 of the system verification, an audit of the meta-learner feature extraction revealed a fundamental rank polarity mismatch. Stacking meta-models require that base model inputs occupy a consistent, monotonic metric space.

Let $\mathbf{s}^{(m)} \in \mathbb{R}^N$ denote the raw predictions produced by model m for a grid of N competitors. Historically, a naive ranking operator was defined as:

$$\text{rank}_{\text{naive}}(\mathbf{s})_i = \text{argsort}(\text{argsort}(-\mathbf{s}))_i + 1 \quad (14)$$

For rankers optimizing utility u_i (XGB, LGBM), Eq. (14) correctly assigns rank 1 to the largest utility ($\max(u) \rightarrow 1$). However, for regression models predicting position y_i directly (RF), the best competitor possesses the minimum predicted value ($y = 1.0$). Applying Eq. (14) directly evaluates $-y$, mapping $y = 1.0 \rightarrow -1.0$ and $y = 20.0 \rightarrow -20.0$, thereby assigning rank 1 to the *slowest* competitor and rank 20 to the *winner*.

This polarity inversion forced the Random Forest predictions to anti-correlate with the gradient-boosted rankers, resulting in negative Spearman rank correlations ($\rho = -0.85$) during out-of-sample backtesting.

To resolve this, we formulated the dual-polarity rank transformation operator $\mathcal{R}(\mathbf{s}, \omega)$ parameterized by polarity indicator $\omega \in \{\text{asc}, \text{desc}\}$:

$$\mathcal{R}(\mathbf{s}, \omega)_i = \begin{cases} \text{argsort}(\text{argsort}(-\mathbf{s}))_i + 1, & \text{if } \omega = \text{desc (higher is better),} \\ \text{argsort}(\text{argsort}(\mathbf{s}))_i + 1, & \text{if } \omega = \text{asc (lower is better).} \end{cases} \quad (15)$$

The rank values are subsequently min-max normalized onto the unit hypercube:

$$\hat{r}_i^{(m)} = \frac{\mathcal{R}(\mathbf{s}^{(m)}, \omega_m)_i - 1}{N - 1} \in [0, 1] \quad (16)$$

where $\omega_{\text{RF}} = \text{asc}$ and $\omega_{\text{XGB}} = \omega_{\text{LGBM}} = \text{desc}$.

3.4 Constrained Ridge Stacking Meta-Learner

The normalized rank vectors $[\tilde{\mathbf{r}}^{(\text{RF})}, \tilde{\mathbf{r}}^{(\text{XGB})}, \tilde{\mathbf{r}}^{(\text{LGBM})}]$ form the input matrix $\mathbf{Z} \in \mathbb{R}^{N \times 3}$ to a Ridge regression meta-learner [7]:

$$\min_{\beta} \|\mathbf{y} - \mathbf{Z}\beta\|_2^2 + \alpha \|\beta\|_2^2 \quad \text{s.t.} \quad \beta_j \geq 0, \quad \sum_{j=1}^3 \beta_j = 1 \quad (17)$$

The ensemble predicted position is then evaluated as $\hat{\mathbf{y}} = \mathbf{Z}\beta^*$. Following the deployment of Eq. (15), the out-of-sample Spearman rank correlation inverted immediately from -0.85 to $+0.830$, with MAE improving to 2.70 on 2024 validation circuits.

4 EMPIRICAL FUEL-CORRECTED TIRE DEGRADATION PHYSICS

Tire degradation represents one of the primary physical determinants of race pace and pit strategy viability [8]. In Free Practice (FP2) long runs, teams assess compound longevity across continuous stints of 10 to 25 laps. However, naively regressing raw telemetry lap times against lap number introduces severe econometric bias due to the confounding effect of vehicle mass burnoff.

4.1 Confounding Fuel Mass Dynamics

During a full Grand Prix distance ($D \approx 305$ km), a modern Formula 1 vehicle consumes approximately 100 kg to 110 kg of fuel across $L_{\text{race}} \approx 50$ –70 laps, yielding an average fuel consumption rate of:

$$\dot{m}_{\text{fuel}} \approx 1.8 \text{ to } 2.0 \text{ kg / lap} \quad (18)$$

Vehicle mass reduction reduces the normal load and inertial resistance through corners and acceleration phases, producing an empirical lap time improvement:

$$\lambda_{\text{fuel}} \approx 0.035 \text{ seconds / lap / kg-burnoff} \quad (19)$$

Consequently, if true physical tire degradation β_{deg} is less than the fuel burnoff advantage ($\beta_{\text{deg}} < \lambda_{\text{fuel}}$), the raw observed lap times $t_{\text{raw}}(l)$ will decrease monotonically over the course of the stint, producing a spurious negative degradation slope ($\beta_{\text{raw}} < 0$). This violates physical thermodynamics and misleads strategy models into anticipating infinite tire durability.

4.2 Fuel-Corrected Formulation

To isolate true tire wear, the engine applies a linear fuel mass correction offset. For a stint comprising laps $l \in \{1, 2, \dots, L_{\text{stint}}\}$:

$$t_{\text{corrected}}(l) = t_{\text{raw}}(l) - \lambda_{\text{fuel}} \cdot (L_{\text{stint}} - l) \quad (20)$$

where $\lambda_{\text{fuel}} = 0.035$ s/lap. Under Eq. (20), lap times are normalized to constant full-fuel equivalent mass.

4.3 Telemetry Filtering and Slope Estimation

Prior to slope estimation, raw telemetry extracted via the FastF1 interface [9] is filtered according to strict green-flag telemetry integrity criteria:

- 1) **In/Out-Lap Elimination:** Stint entry (pit exit) and in-laps (pit entry) are discarded.
- 2) **Traffic and Yellow Flag Filtering:** Laps with track status flags $\neq 1$ (Safety Car, VSC, Yellow) or exceeding the session personal best by more than 7% ($t_{\text{raw}} > 1.07 \cdot t_{\text{best}}$) are pruned.
- 3) **Minimum Stint Length:** Only stints with $N_{\text{clean}} \geq 5$ consecutive valid laps are accepted.

The physical tire degradation gradient β_{deg} is estimated via Ordinary Least Squares (OLS):

$$\beta_{\text{deg}} = \frac{\sum_{l=1}^L (l - \bar{l})(t_{\text{corrected}}(l) - \bar{t}_{\text{corrected}})}{\sum_{l=1}^L (l - \bar{l})^2} \quad (21)$$

Following Eq. (21), empirical tests across Bahrain and Barcelona telemetry confirm that $\beta_{\text{deg}} > 0$ across all dry compounds (typically 0.020–0.085 s/lap for C3 Medium and 0.050–0.140 s/lap for C4/C5 Soft compounds).

4.4 Offline Scikit-Learn Regression Fallback

When external telemetry APIs are unreachable or when operating in air-gapped evaluation environments, the engine activates the `SklearnTireDegradationModel`. This fallback module fits a Multi-Layer Perceptron (MLP) regressor to historical Pirelli compound wear curves conditioned on:

$$\hat{\beta}_{\text{deg}} = f_{\theta}(\text{compound}, T_{\text{track}}, L_{\text{stint}}, \text{abrasion_index}) \quad (22)$$

ensuring deterministic degradation gradients without external network dependencies.

5 BAYESIAN POINT CALIBRATION AND ZERO-INFLATED DNF PENALTIES

In fantasy sports, winning portfolios are determined not by raw finishing positions, but by the expected value and variance of rule-based fantasy points $\text{Pts} \in \mathbb{R}$. Formula 1 Fantasy employs a highly non-linear, multi-attribute scoring schedule that penalizes retirements severely.

5.1 Official Scoring Function and Asymmetric Payoffs

For driver i with starting grid g_i and finishing position p_i , the deterministic fantasy point reward is:

$$\begin{aligned} \text{Pts}(g_i, p_i) = & \text{Pts}_{\text{race}}(p_i) + \text{Pts}_{\text{quali}}(g_i) \\ & + \text{clamp}(g_i - p_i, -10, +10) \\ & + \text{Bonus}_{\text{teammate}} + \text{Bonus}_{\text{FL}} + \text{Bonus}_{\text{DOTD}} \end{aligned} \quad (23)$$

where $\text{Pts}_{\text{race}}(p_i)$ follows the FIA World Championship scale (25, 18, 15, $\dots$), positions gained/lost award ± 1 point per position delta (bounded within $[-10, +10]$), and defeating a teammate awards +3 points.

Critically, a Did Not Finish (DNF) event incurs a direct penalty:

$$\text{Pts}_{\text{DNF}} = -15.0 \text{ points} \quad (24)$$

5.2 The Zero-Inflation Retirement Bias and Fix

Prior to Phase 2 of our verification audit, standard prediction engines suffered from *zero-inflation optimistic bias*. Specifically, when computing expected points $\mathbb{E}[\text{Pts}_i]$, models either omitted DNF outcomes or mapped retirements to $p_i = 20$ (which yielded 0 points plus position-loss penalties). Consequently, high-risk backmarker drivers with 25% mechanical retirement probability were assigned positive expected yields ($\mathbb{E} \approx +6.5$ pts), triggering disastrous algorithmic allocations in low-budget optimizer rosters.

To establish rigorous calibration, we formulated a zero-inflated two-component mixture distribution [10]. Let $Z_i \in \{0, 1\}$ denote the binary retirement indicator ($Z_i = 1$ denotes DNF) with probability $P_{\text{dnf},i}$. The true calibrated expected fantasy point yield is:

$$\mathbb{E}[\text{Pts}_i] = (1 - P_{\text{dnf},i}) \cdot \mathbb{E}[\text{Pts}_i \mid Z_i = 0] + P_{\text{dnf},i} \cdot \text{DNF_PENALTY} \quad (25)$$

The retirement probability $P_{\text{dnf},i}$ is modeled via empirical Bayes pooling across driver, chassis, and circuit risk factors:

$$P_{\text{dnf},i} = \text{clamp}(0.40 \cdot \theta_{\text{driver},i} + 0.40 \cdot \theta_{\text{team},c} + 0.20 \cdot \theta_{\text{circuit}}, 0.02, 0.45) \quad (26)$$

where $\theta_{\text{driver},i}$ and $\theta_{\text{team},c}$ represent historical 2-season rolling DNF rates, and θ_{circuit} represents track incident frequency (e.g., Singapore, Baku, or Monaco).

Under Eq. (25), volatile drivers with elevated retirement rates appropriately register negative or depressed expected yields ($\mathbb{E}[\text{Pts}] \approx -2.0$ to $+1.5$), protecting the downstream optimizer from fragile assets.

5.3 Conjugate Beta-Binomial Teammate Dominance

The teammate bonus (+3 pts) represents an important source of alpha. For teammates (A, B), the probability $P(A > B)$ is modeled as a Beta-Binomial conjugate posterior:

$$P(A > B \mid \mathcal{D}) = \frac{\alpha_0 + k_A}{\alpha_0 + \beta_0 + N_{AB}} \quad (27)$$

where $\alpha_0 = \beta_0 = 2$ represents an uninformative prior, N_{AB} denotes completed head-to-head races, and k_A denotes victories by driver A .

6 CORRELATED LAP-BY-LAP MONTE CARLO SIMULATION PHYSICS

While point expectation estimators provide first-order guidance, optimal portfolio management under asymmetric risk requires modeling higher-order distribution moments, skewness, and tail risk. The engine incorporates a stochastic, lap-by-lap Monte Carlo simulation engine executing $M = 10,000$ virtual Grands Prix per circuit [8].

6.1 Consistency-Modulated Stochastic Pace Dynamics

On lap $l \in \{1, 2, \dots, L\}$, the simulated lap time $t_{i,l}$ of driver i is governed by:

$$t_{i,l} = t_{\text{base},i} + \beta_{\text{deg},i} \cdot l_{\text{stint}} - \lambda_{\text{fuel}} \cdot (L - l) + \epsilon_{i,l} \quad (28)$$

where $t_{\text{base},i}$ is derived from the ML ensemble predicted position $\hat{y}_i$, and $\epsilon_{i,l} \sim \mathcal{N}(0, \sigma_i^2)$ represents stochastic driving noise.

In Phase 8, we integrated a consistency-scaled variance model inspired by real telemetry variability:

$$\sigma_i = \sigma_{\text{base}} \cdot \left(1.0 - 0.50 \cdot \frac{C_i}{100}\right) \quad (29)$$

where $\sigma_{\text{base}} = 0.150$ seconds, and $C_i \in [50, 100]$ denotes the empirical consistency index computed from clean lap-time coefficients of variation across prior races. For world champions ($C \approx 95$), $\sigma_i \approx 0.078$ s, whereas erratic rookies ($C \approx 60$) exhibit $\sigma_i \approx 0.105$ s.

6.2 Safety Car Incidents and Midfield Jump Clamping

Safety Cars compress the field and induce opportunistic pit stops, creating substantial variance. On each lap l , a Bernoulli trial determines whether an SC or VSC occurs:

$$P(\text{SC}_l) = \frac{P_{\text{SC}}(\text{circuit})}{L_{\text{race}}} \cdot \psi(l) \quad (30)$$

where $\psi(l)$ is the first-lap incident hazard modifier ($\psi(1) = 2.50$, $\psi(l) = 1.0$ for $l > 1$).

During Phase 3 of our audit, an analysis of Monte Carlo distributions revealed an unphysical simulation artifact: under neutral SC pit windows, mid-pack cars (running P_8 to P_{15}) occasionally jumped directly to P_1 or P_2 due to unconstrained random shuffle deltas. In physical Formula 1 racing, unlapping regulations, pit exit timing, and aerodynamic wake restrict positions gained under SC to tactical pit delta advantages.

To enforce physical realism, we instituted a midfield position jump clamp. For a car in running position $p_{i,l}$ during an SC event:

$$p_{i,l+1} = p_{i,l} + \text{clamp}(\delta_{\text{pit},i}, -\min(3, p_{i,l} - 1), +4) \quad (31)$$

Eq. (31) restricts maximum forward gain under a single SC intervention to ≤ 3 positions, preventing unphysical podium finishes for midfield vehicles while fully preserving realistic strategic gains.

6.3 Constructor Expected Fantasy Points Alignment

Constructors score points as the sum of their two drivers' fantasy race points plus pit stop efficiency bonuses ($\text{Bonus}_{\text{pit}}$):

$$\mathbb{E}[\text{Pts}_c] = \sum_{i \in \mathcal{D}_c} (\mathbb{E}[\text{Pts}_i] - \mathbb{E}[\text{Bonus}_{\text{DOTD},i}]) + \mathbb{E}[\text{Bonus}_{\text{pit},c}] \quad (32)$$

Note that Driver of the Day (DOTD = +10 pts) accrues exclusively to the driver and is strictly excluded from constructor totals, an invariant verified in our Phase 3 unit tests.

7 CONSTRAINED MIXED-INTEGER LINEAR PROGRAMMING TEAM OPTIMIZER

Selecting an optimal fantasy lineup requires solving a constrained combinatorial knapsack problem. Because fantasy points are strictly additive across selected assets, the global portfolio optimization can be formulated as a Mixed-Integer Linear Program (MILP) solved via branch-and-cut algorithms.

7.1 Mathematical Program Formulation

Let $\mathcal{D}$ denote the set of $N = 22$ drivers and $\mathcal{C}$ denote the set of $K = 11$ constructors. Define binary decision variables:

$$x_i \in \{0, 1\}, \quad \forall i \in \mathcal{D} \quad (\text{driver } i \text{ selected}) \quad (33)$$

$$y_j \in \{0, 1\}, \quad \forall j \in \mathcal{C} \quad (\text{constructor } j \text{ selected}) \quad (34)$$

$$t_i \in \{0, 1\}, \quad \forall i \in \mathcal{D} \quad (\text{driver } i \text{ designated as Turbo Driver}) \quad (35)$$

Let c_i and c_j denote asset market prices, and let $\mathbb{E}[\text{Pts}_i]$ and $\mathbb{E}[\text{Pts}_j]$ denote the corresponding expected point yields. The objective is to maximize total expected portfolio yield minus transfer penalties:

$$\max_{\mathbf{x}, \mathbf{y}, \mathbf{t}} \sum_{i \in \mathcal{D}} \mathbb{E}[\text{Pts}_i] x_i + \sum_{j \in \mathcal{C}} \mathbb{E}[\text{Pts}_j] y_j + \sum_{i \in \mathcal{D}} \mathbb{E}[\text{Pts}_i] t_i - 10 \cdot \Delta_{\text{transfers}} \quad (36)$$

subject to the canonical roster constraints:

$$\sum_{i \in \mathcal{D}} x_i = 5 \quad (\text{roster size: exactly 5 drivers}) \quad (37)$$

$$\sum_{j \in \mathcal{C}} y_j = 2 \quad (\text{roster size: exactly 2 constructors}) \quad (38)$$

$$\sum_{i \in \mathcal{D}} c_i x_i + \sum_{j \in \mathcal{C}} c_j y_j \leq B \quad (\text{budget constraint}) \quad (39)$$

7.2 Turbo Driver Coupling and Price Thresholds

The Turbo Driver chip doubles the points scored by a single designated driver ($2 \times$ multiplier). In the linear objective of Eq. (36), this is represented by adding $\mathbb{E}[\text{Pts}_i] t_i$. The Turbo selection is governed by coupling and eligibility constraints:

$$\sum_{i \in \mathcal{D}} t_i = 1 \quad (\text{exactly one Turbo driver}) \quad (40)$$

$$t_i \leq x_i, \quad \forall i \in \mathcal{D} \quad (\text{must be selected in roster}) \quad (41)$$

$$c_i t_i \leq 20.0, \quad \forall i \in \mathcal{D} \quad (\text{maximum Turbo price ceiling}) \quad (42)$$

Under Eq. (42), premium drivers priced above \$20.0M (such as Max Verstappen or Lando Norris) are mathematically precluded from Turbo designation.

7.3 The 2026 Team Cap Constraint and Roster Invariants

A critical finding during Phase 6 of our audit involved the enforcement of the 2-asset-per-constructor constraint. Under 2026 Formula 1 Fantasy sporting regulations, a participant cannot hold more than 2 assets from the same constructor (e.g., selecting both Ferrari drivers precludes selecting Ferrari as a constructor).

Historically, the constraint was hardcoded to 10 static 2025 constructors, omitting the 11th entry (Cadillac F1) and permitting illegal triples in dynamic roster tests. We implemented a dynamic constraint mapping:

$$\sum_{i \in \text{Lineup}(k)} x_i + y_k \leq 2, \quad \forall k \in \mathcal{C} \quad (43)$$

where $\text{Lineup}(k) = \{i \in \mathcal{D} \mid \text{team}(i) = k\}$ and $|\mathcal{C}| = 11$.

The resulting integer program contains $22 + 11 + 22 = 55$ binary decision variables and $1 + 1 + 1 + 1 + 22 + 22 +$

$11 = 59$ linear inequalities. The problem is solved in under 25 milliseconds using the COIN-OR Branch and Cut (CBC) solver via the PuLP interface.

8 EMPIRICAL VALIDATION AND MULTI-SEASON BENCHMARKS

To rigorously assess the predictive power of the mathematical architecture, we conduct extensive walk-forward multi-season backtesting across completed Formula 1 championships (2021–2025) and simulated 2026 calendar configurations.

8.1 Multi-Race Lookahead and Grid Isolation

Fantasy roster transfers carry transaction frictions: teams receive only two free transfers per Grand Prix, with additional substitutions penalized at -10 points each. A purely myopic optimizer that maximizes single-race expected value $\mathbb{E}[\text{Pts}_r]$ frequently acquires high-performing assets tailored to a unique circuit profile (e.g., Monza or Monaco) that underperform immediately thereafter.

To eliminate myopic instability, the engine implements a multi-race lookahead expectation over horizon $H = 3$ races with exponential temporal discounting [8]:

$$\mathbb{E}[\text{Pts}_a] = \frac{\sum_{k=0}^{H-1} \gamma^k \cdot \mathbb{E}[\text{Pts}_{r+k}(a)]}{\sum_{k=0}^{H-1} \gamma^k} \quad (44)$$

where $\gamma = 0.75$ represents the inter-round decay factor ($w_0 = 1.000$, $w_1 = 0.750$, $w_2 = 0.5625$).

A critical bug resolved during Phase 7 was *grid override leakage*: manual or penalty grid displacements specified for round r previously persisted into the predictor context for future rounds $r + 1$ and $r + 2$. The hardened pipeline strictly isolates grid overrides to round r , enforcing clean synthetic grids for subsequent calendar rounds.

8.2 Empirical Backtest Metrics

System performance is evaluated using five standard statistical metrics:

- **Mean Absolute Error (MAE):** $\frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$.
- **Root Mean Squared Error (RMSE):** $\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$.
- **Spearman Rank Correlation (ρ):** Non-parametric monotonicity metric measuring ranking fidelity.
- **Top- K Hit Rate ($K \in \{3, 5\}$):** Proportion of true top- K finishers correctly predicted inside the top- K .
- **Winner Accuracy:** Binary indicator of whether the true race winner was identified in P_1 .

8.3 Ablation Analysis

To demonstrate the marginal contribution of each architectural enhancement identified during our 10-phase verification plan, Table 2 reports an ablation study across five model configurations.

The ablation results highlight the catastrophic consequence of Configuration 3: without dual-polarity rank transformation (Eq. (15)), the regression model inverts rank directions, yielding a severe negative rank correlation

TABLE 1
Predictive Performance on 2024 Season Benchmark Circuits.

Evaluation Metric	Observed Benchmark Value
Race Finishing MAE	2.700 positions
Race Finishing RMSE	3.362 positions
Race Spearman Rank Correlation (ρ)	+0.830
Qualifying Finishing MAE	2.900 positions
Qualifying Spearman Rank Correlation	+0.777
Race Winner Accuracy	100.0%
Top-3 Podium Hit Rate	66.7% (2 / 3)
Top-5 Hit Rate	80.0% (4 / 5)

TABLE 2
Ablation Analysis of Model Enhancements.

Model Configuration	Race MAE	Spearman ρ
1. Naive Starting Grid Baseline	4.85	+0.451
2. Standalone Glicko-2 Elo	3.60	+0.624
3. Uncorrected ML Ensemble (Inverted RF)	6.95	-0.852
4. ML Ensemble with Dual-Polarity Stacking	2.70	+0.830
5. Full Physical Engine (Fuel Deg + Consistency MC)	2.55	+0.854

($\rho = -0.852$). Restoring mathematical rank alignment (Configuration 4) and incorporating physical tire degradation and consistency-modulated variance (Configuration 5) delivers state-of-the-art predictive fidelity (MAE = 2.55, $\rho = +0.854$).

8.4 Monotonic Information Gain Across Weekend Stages

During a Grand Prix weekend, prediction uncertainty contracts as empirical telemetry accumulates. We evaluated stage-by-stage performance across three distinct operational modes:

- 1) **Pre-Practice Mode (Thursday):** Incorporates Glicko-2 ratings, historical circuit profiles, and season EWMA form (MAE = 3.20, $\rho = 0.72$).
- 2) **Post-Practice Mode (Friday/Saturday):** Integrates FP2 fuel-corrected tire wear slopes β_{deg} and practice pace deltas (MAE = 2.85, $\rho = 0.79$).
- 3) **Post-Qualifying Mode (Saturday/Sunday):** Ingests actual starting grids, qualifying sector deltas, and grid penalties (MAE = 2.55, $\rho = 0.854$).

The monotonic reduction in MAE and expansion in rank correlation demonstrates that the pipeline effectively assimilates new real-world information without regression.

9 SOFTWARE VERIFICATION, SECURITY GATES, AND CONCLUSION

Mission-critical predictive pipelines deployed in competitive gaming and financial markets demand strict software engineering discipline. To ensure robustness, maintainability, and reproducibility, the F1 Prediction Engine incorporates a comprehensive quality assurance architecture.

9.1 Verification Test Suite and Offline Mocking

The system is protected by a continuous verification suite spanning over 100 unit and contract tests located in `engine/tools/tests/`:

- **Mathematical Contract Tests (`test_contracts.py`):** Validates scoring parity, Glicko-2 volatility convergence bounds, Monte Carlo midfield jump clamps, lookahead temporal discounting, and the 2026 two-asset team cap constraint.
- **End-to-End Pipeline Smoke Tests (`test_pipeline.py`, `test_serialize.py`):** Verifies data flow and strict JSON serializability of event payloads transmitted across WebSocket connections to the user interface.
- **Benchmark Validation Tests (`test_backtest_benchmarks.py`):** Guarantees that historical backtest metric computations conform to standard mathematical definitions without numerical drift.
- **Offline Test Isolation:** All integration tests isolate network scrapers via deterministic mock fixtures (`unittest.mock.patch`), ensuring that automated test discovery and CI test execution run reliably in isolated environments without network latency or headless browser timeouts.

9.2 Nuclear Secret and PII Scanning Architecture

To eliminate the risk of accidental credential exposure or privacy violations, the repository enforces a pre-commit and pre-push security gate via `engine/tools/secret_scanner.py`. The scanner executes regular expression matching across:

- 1) Active working directory files and tracked git entities.
- 2) Staged diff payloads prior to commit finalization.
- 3) Full historical git commit logs across all repository revisions.

The scanner enforces zero tolerance for active API keys (e.g., OpenWeatherMap, FastF1, Jolpica tokens), authentication cookies, raw session tokens, private cryptographic keys, and Personal Identifying Information (PII). Any violation aborts the build and blocks git operations.

9.3 Conclusion and Future Research Directions

This monograph presented the mathematical foundations, architectural hardening, and empirical verification of the open-source F1 Prediction Engine. By resolving foundational rank polarity mismatches in the stacking meta-learner, calibrating zero-inflated DNF penalties, implementing fuel-corrected empirical tire degradation, and enforcing dynamic 2026 regulations within a mixed-integer linear programming optimizer, the engine delivers robust, state-of-the-art predictive performance across multi-season backtests.

Future research directions include:

- **Neural Ordinary Differential Equations (Neural ODEs):** Modeling continuous within-stint lap-time degradation and tire surface temperature transients via continuous-depth neural networks.
- **Game-Theoretic Bayesian Dynamic Games:** Simulating multi-agent strategic interactions (undercut and overcut decisions) under live track telemetry using imperfect-information game trees.

- **Reinforcement Learning for Multi-Round Chips:** Training deep Q-networks (DQN) to optimize the dynamic activation timing of single-use fantasy chips (e.g., Triple Booster, Wildcard, No Negative) across a 24-race calendar.

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